PHYS1408 Practice Exam 3-3 — Answer Key

Worked solutions with course-material citations

Every numeric answer was computed and checked in `_scripts/build_practice-exam_3_3.py`. Constants are the ones printed on the 2015 Exam 3: $G = 6.67 \times 10^{-11}$, $M_{earth} = 5.98 \times 10^{24}$ kg, $R_{earth} = 6.37 \times 10^{6}$ m, and $g = 9.8$ m/s^2.

Sources. "Ch 12 notes p. 6/10" means page 6 of the professor's *Chapter 12 Statics* notes (he numbers his pages "6/10"). "2015 Exam 3 #2" is question 2 of the previous exam in `Professors-Notes/exam 3-4.pdf`.

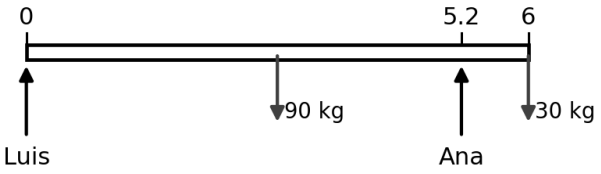
Practice Exam 3-3 mixes the 2015 Exam 3 types (sofa with a load on the end, the climbing man, two cords, the wire, the jello block, g at an altitude, satellite period, launch speed, binary stars, the equal-pull point) with notes examples: the angle needed to reach the top of the ladder, the diving board, the seesaw, the rocket height, and escape speed. Question 2 is a **maximum**: pick the largest choice that still works. Every question is a procedure worked in the notes or on the 2015 exam.

Q	Ans	Answer	Type	Topic (source)	Notecard
1	**e**	848 N	Calc	Two people carry a sofa (Ch 12 p. 4–5; 2015 #1)	Sofa / beam
2	**a**	5.5 m	Max	Man climbing a ladder, x_{max} (Ch 12 p. 8; 2015 #2)	Ladder
3	**e**	61.2°	Calc	Ladder angle to reach the top (Ch 12 p. 9)	Ladder
4	**d**	299 N	Calc	Light hung by two cords (2015 #3; Ch 12 p. 1)	Two cords
5	**d**	13.0 mm	Calc	Young's modulus, wire stretch (Ch 12 p. 10; 2015 #5)	Moduli
6	**b**	2.35 mm	Calc	Young's modulus, compressed block (2015 #4; Ch 12 p. 10)	Moduli

Q	Ans	Answer	Type	Topic (source)	Notecard
7	**c**	7,350 N	Calc	Diving board, support B (Ch 12 p. 3)	Diving board
8	**b**	0.450 m	Calc	Seesaw balance (Ch 12 p. 4)	Seesaw
9	**a**	8.97 m/s²	Calc	g at an altitude (Ch 13 p. 1–2; 2015 #12)	Gravity
10	**e**	101 min	Calc	Satellite period (Ch 13 p. 2; 2015 #10)	Orbit
11	**a**	493 km	Calc	Height reached from the launch speed (Ch 13 p. 7–8)	Rocket
12	**b**	5.47 km/s	Calc	Launch speed to reach a height (Ch 13 p. 7–8; 2015 #9)	Rocket
13	**c**	5.01 km/s	Calc	Escape speed (Ch 13 p. 9)	Escape
14	**d**	89.9%	Calc	Equal-pull point (2015 #13; Ch 13 p. 1)	Equal pull
15	**c**	87.2 hours	Calc	Binary stars, period (Ch 13 p. 4; 2015 #11)	Binary stars

1. Two people carry a sofa — e) 848 N

Source: Ch 12 notes p. 4/10 (two men carry a sofa: $\sum \tau_{Jim}$, $\sum \tau_{Sam}$) and p. 5/10 (20 kg kid set on the sofa 1 m from Jim).
Previous-exam match: 2015 Exam 3 #1 (sofa + box of books on Sarah's end), new numbers
Notecard: Sofa / beam
What it tests: torque balance about one carrier to find the other carrier's force.



Step 1: pick the pivot at Luis's hands. A force applied at the pivot has $r = 0$, so its torque is zero and it drops out of the equation. The notes do this on p. 3/10 ("A is eliminated").

Step 2: torque balance about Luis. Every force here is vertical and the sofa is horizontal, so each torque is $\tau = rF$, with r measured from Luis. The sofa's weight acts at its center, $L/2$. Ana's upward push is counterclockwise; the weights are clockwise.

$$\sum \tau_{Luis} = 0: \quad x_{Ana}\, F_{Ana} = \tfrac{L}{2}\, Mg + x_1\, m_1 g$$

Step 3: plug in ($x_{Ana} = 5.2$ m):

$$5.2 \text{ m} \times F_{Ana} = \underbrace{3 \text{ m}}_{\text{sofa center}}\ \underbrace{(90)(9.8)}_{\text{sofa weight}} + \underbrace{6 \text{ m}}_{\text{crate position}}\ \underbrace{(30)(9.8)}_{\text{crate weight}}$$

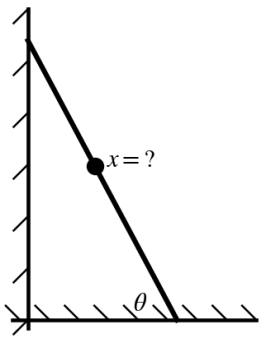
$$F_{Ana} = \frac{4{,}410 \text{ N m}}{5.2 \text{ m}} = 848.1 \text{ N}$$

Why the other choices are wrong:

- a) 328 N: is the force on Luis, the other person.
- b) 509 N: leaves out the added load.
- c) 588 N: splits the total weight evenly; that is only true with nothing extra and both at the ends.
- d) 735 N: uses 6 m (the full length) as Ana's lever arm instead of 5.2 m.

2. Man climbing a ladder, x_{max} — a) 5.5 m

Source: Ch 12 notes p. 8/10 (leaning ladder with a man climbing it, $x = \frac{L}{M}(\mu_s(M+m)\tan\theta - \frac{m}{2}) = 8.8$ m).
Previous-exam match: 2015 Exam 3 #2 (150 kg man, 20 kg ladder, 71°, $\mu = 0.25$, 5 m), new numbers
Notecard: Ladder
What it tests: ladder torque balance with a climber, then the maximum-answer rule.



Step 1: forces (notes p. 8/10). $\sum F_y$: $N = (M+m)g$. $\sum F_x$: $N_w = F_f$, and at the limit $F_f = \mu_s N = \mu_s (M+m)g$.

Step 2: torque about the base. Ladder weight at $L/2$ (arm $\frac{L}{2}\cos\theta$), man at x (arm $x\cos\theta$), wall force at the top (arm $L\sin\theta$):

$$\tfrac{L}{2}\cos\theta\, mg + x\cos\theta\, Mg - L\sin\theta\, \mu_s (M+m)g = 0$$

Cancel g, divide by $\cos\theta$:

$$x = \frac{L}{M}\Big(\mu_s (M+m)\tan\theta - \frac{m}{2}\Big)$$

Step 3: plug in ($\tan 62^\circ = 1.881$):

$$x = \frac{\overbrace{8}^{L}}{\underbrace{110}_{M}}\Big(\underbrace{(0.35)}_{\mu_s}\,\underbrace{(140)}_{M+m}(1.881) - \underbrace{\tfrac{30}{2}}_{m/2}\Big) = 5.61 \text{ m}$$

Step 4: it's a maximum. He can go up to 5.61 m; any farther and the ladder slips. The largest choice that still works is **5.5 m**. (On the 2015 exam, $x = 3.78$ m and the answer was b) 3.5 m.)

Why the other choices are wrong:

- b) 6.0 m: is more than the maximum, so it fails.
- c) 6.5 m: is more than the maximum, so it fails.
- d) 7.0 m: is more than the maximum, so it fails.
- e) 7.5 m: is more than the maximum, so it fails.

3. Ladder angle to reach the top — e) 61.2°

Source: Ch 12 notes p. 9/10 ("He can climb to $x = L$, what θ is minimum?" $\theta_{min} = \tan^{-1}\frac{M+m/2}{(M+m)\mu_s} = 67.6°$).
Previous-exam match: none (lecture-notes example)
Notecard: Ladder
What it tests: the notes p. 9 procedure: solve for θ with $x = L$.

Step 1: start from the climbing result (notes p. 8/10), $xM = \mu_s L(M+m)\tan\theta - \frac{mL}{2}$, and set $x = L$ (notes p. 9/10, "He can climb to $x = L$, what θ is minimum?"):

$$LM + L\frac{m}{2} = \mu_s L(M+m)\tan\theta \;\Rightarrow\; \tan\theta = \frac{M+m/2}{(M+m)\mu_s}$$

L cancels, so the ladder's length doesn't matter.

Step 2: plug in:

$$\theta_{min} = \tan^{-1}\frac{\overbrace{90+10}^{M+m/2}}{\underbrace{(110)}_{M+m}\underbrace{(0.5)}_{\mu_s}} = \tan^{-1}(1.818) = 61.2°$$

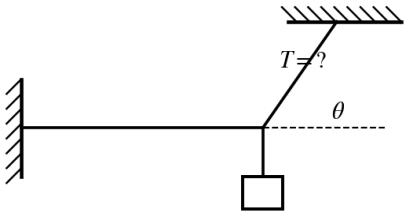
Notes numbers ($M = 150$, $m = 10$, $\mu_s = 0.4$): $\tan^{-1}(155/64) = 67.6°$.

Why the other choices are wrong:

- a) 14.7°: No standard mistake gives this value; it is a spacing choice.
- b) 20.6°: No standard mistake gives this value; it is a spacing choice.
- c) 28.8°: is the angle from the wall, not from the floor.
- d) 45.0°: is θ_{min} for the empty ladder (p. 7/10); it ignores the man.

4. Light hung by two cords — d) 299 N

Source: 2015 Exam 3 #3, solved with $\sum F_x = \sum F_y = 0$ from Ch 12 notes p. 1/10 ("Static means not moving").
Previous-exam match: 2015 Exam 3 #3 (20 kg light, 40° with the ceiling: 305 N → b) 300 N), new numbers
Notecard: Two cords
What it tests: $\sum F_y = 0$ on the knot: $T\sin\theta = mg$.



Step 1: forces on the knot (static, so $\sum F_x = \sum F_y = 0$, notes p. 1/10). Ceiling cord T at θ above the horizontal; horizontal cord T_h; the light's weight mg hangs straight down.

Step 2: $\sum F_y = 0$. Only the ceiling cord has a vertical part, $T\sin\theta$:

$$T\sin\theta = mg \;\Rightarrow\; T = \frac{mg}{\sin\theta} = \frac{\underbrace{(25)(9.8)}_{mg=245.0\text{ N}}}{\sin 55^\circ} = 299.1\text{ N}$$

Why the other choices are wrong:

- a) 172 N: is the tension in the horizontal cord.
- b) 201 N: multiplies by $\sin\theta$ instead of dividing.
- c) 245 N: is just the light's weight.
- e) 427 N: uses $\cos\theta$: that would be the angle from the vertical.

5. Young's modulus, wire stretch — d) 13.0 mm

Source: Ch 12 notes p. 10/10 (Young's modulus $Y = \frac{F/A}{\Delta\ell/\ell}$).
Previous-exam match: 2015 Exam 3 #5 (500 kg, 4.00 m, 0.001 m diameter, 8×10^{10}: 0.312 m $\rightarrow$ e) 0.3 m), new numbers
Notecard: Moduli
What it tests: tensile stress/strain with a circular cross-section ($r = d/2$).

Step 1: the formula (Ch 12 notes p. 10/10):

$$Y = \frac{\text{tensile stress}}{\text{tensile strain}} = \frac{F/A}{\Delta\ell/\ell}$$

Step 2: the pieces. The force is the load's weight, and the area is the wire's circular cross-section (radius = half the diameter):

$$F = mg = (200)(9.8) = 1{,}960\text{ N}$$

$$A = \pi r^2 = \pi\Big(\underbrace{\tfrac{0.002}{2}}_{r}\Big)^2 = 3.142 \times 10^{-6}\ \text{m}^2$$

Step 3: solve for the stretch:

$$\Delta\ell = \frac{F\,\ell}{A\,Y} = \frac{\underbrace{(1{,}960)}_{F}\,\underbrace{(2.5)}_{\ell}}{\underbrace{(3.142 \times 10^{-6})}_{A}\,\underbrace{(1.2 \times 10^{11})}_{Y}} = 0.0130\ \text{m}$$

Why the other choices are wrong:

- a) 1.33 mm: forgets g (uses the mass instead of the weight).
- b) 3.25 mm: uses the diameter as the radius in $A = \pi r^2$.
- c) 10.2 mm: uses $A = d^2$ (no π, no half).
- e) 52.0 mm: uses $r = d/4$ (halves the radius twice).

6. Young's modulus, compressed block — b) 2.35 mm

Source: 2015 Exam 3 #4, solved with Young's modulus from Ch 12 notes p. 10/10.
Previous-exam match: 2015 Exam 3 #4 (jello 3 m × 4 m × 0.1 m, 500 kg: 1.36 mm → e) 1.3 mm), new numbers
Notecard: Moduli
What it tests: compressive stress on the top face, strain = $\Delta h/h$.

Step 1: the formula (Ch 12 notes p. 10/10), with compression instead of stretch:

$$Y = \frac{F/A}{\Delta h/h} \;\Rightarrow\; \Delta h = \frac{F\,h}{A\,Y}$$

Step 2: the pieces. The mass pushes down on the **top** face, so A = width × length. h is the original height.

$$F = (150)(9.8) = 1{,}470\ \text{N}$$

$$A = (0.5)(0.5) = 0.2500\ \text{m}^2$$

Step 3: plug in:

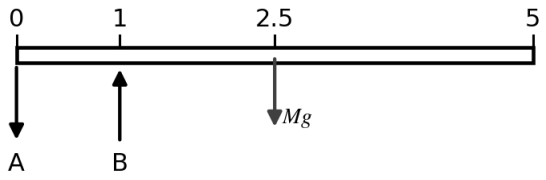
$$\Delta h = \frac{\underbrace{(1{,}470)}_{F}\,\underbrace{(0.2)}_{h}}{\underbrace{(0.2500)}_{\text{top area}}\,\underbrace{(5\times10^5)}_{Y}} = 0.00235 \text{ m} = 2.35 \text{ mm}$$

Why the other choices are wrong:

- a) 0.240 mm: forgets g.
- c) 5.88 mm: uses a side face ($w \times h$) as the area.
- d) 11.8 mm: finds the strain and stops (forgets to multiply by the height).
- e) 16.5 mm: No standard mistake gives this value; it is a spacing choice.

7. Diving board, support B — c) 7,350 N

Source: Ch 12 notes p. 3/10 (diving board, $L = 10$ m, $M = 1000$ kg: $B = 5Mg$, $A = 4Mg$).
Previous-exam match: none (lecture-notes example)
Notecard: Diving board
What it tests: choosing the pivot to eliminate one support, then $\sum F_y = 0$ (A pulls down).



Step 1: draw the forces (notes p. 3/10). The board's weight Mg acts at its center, $L/2 = 2.5$ m. B pushes **up**. A must pull **down**, or the board would tip off B.

Step 2: torque about A removes A ($r = 0$):

$$\sum \tau_A = 0: \quad a\,B - \tfrac{L}{2}\,Mg = 0$$

$$B = \frac{L/2}{a}\,Mg = \frac{\overbrace{2.5 \text{ m}}^{L/2}}{\underbrace{1 \text{ m}}_{\text{A to B}}}\,\underbrace{(300)(9.8)}_{Mg=2{,}940 \text{ N}} = 7{,}350 \text{ N}$$

Step 3: $\sum F_y = 0$ with B up and A, Mg down:

$$B - A - Mg = 0 \;\Rightarrow\; A = B - Mg = 7{,}350 - 2{,}940 = 4{,}410 \text{ N}$$

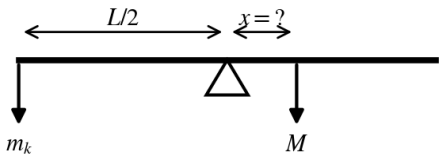
In the notes ($L = 10$ m, $a = 1$ m) this gives $B = 5Mg$ and $A = 4Mg$.

Why the other choices are wrong:

- a) 2,940 N: is just the board's weight Mg.
- b) 4,410 N: is the force from support A, not B.
- d) 10,300 N: adds Mg instead of subtracting it.
- e) 14,700 N: puts the board's weight at the far end (L) instead of its center ($L/2$).

8. Seesaw balance — b) 0.450 m

Source: Ch 12 notes p. 4/10 (seesaw, $m_k = 20$ kg, $M = 100$ kg, $x = L/10$).
Previous-exam match: none (lecture-notes example)
Notecard: Seesaw
What it tests: torque balance about the pivot; the seesaw's own mass drops out.



Step 1: torque about the center (pivot). The seesaw's own weight acts at the center, so its lever arm is zero. That's why the notes say its mass "don't care!" The kid is $L/2$ from the pivot; the adult is x from it.

$$\sum \tau_{center} = 0: \quad \tfrac{L}{2}\, m_k g - x\, M_T g = 0 \;\Rightarrow\; x = \frac{L\, m_k}{2 M_T}$$

g cancels.

Step 2: plug in:

$$x = \frac{\underbrace{3.6 \text{ m}}_{\text{length}}\ \underbrace{18 \text{ kg}}_{\text{kid}}}{2\ \underbrace{72 \text{ kg}}_{\text{adult}}} = 0.450 \text{ m}$$

In the notes, $m_k = 20$ kg and $M_T = 100$ kg give $x = L/10$.

Why the other choices are wrong:

- a) 0.321 m: No standard mistake gives this value; it is a spacing choice.
- c) 0.900 m: forgets that the kid's lever arm is $L/2$, not L.
- d) 1.35 m: measures from the end of the seesaw instead of from the center.
- e) 2.25 m: adds the half-length.

9. g at an altitude — a) 8.97 m/s²

Source: Ch 13 notes p. 1/10 ($g = GM_E/R_E^2 = 9.81$ m/s²) and p. 2/10 ($r = R + h$).
Previous-exam match: 2015 Exam 3 #12 (750 km: 7.87 m/s² → b) 8 m/s²), new numbers
Notecard: Gravity
What it tests: $g = GM/r^2$ with r measured from Earth's center.

Step 1: the formula (notes p. 1/10): $mg = \dfrac{GMm}{R^2}$, so $g = \dfrac{GM}{R^2}$ at the surface. Above the surface the distance is measured from Earth's **center**: $r = R + h$ (notes p. 2/10).

$$g = \frac{GM_{earth}}{(R+h)^2}$$

Step 2: distance from the center:

$$r = \underbrace{6.37 \times 10^6}_{R_{earth}} + \underbrace{3 \times 10^5}_{h \text{ in m}} = 6.670 \times 10^6 \text{ m}$$

Step 3: plug in:

$$g = \frac{\underbrace{(6.67 \times 10^{-11})}_{G}\underbrace{(5.98 \times 10^{24})}_{M_{earth}}}{(6.670 \times 10^6)^2} = 8.97 \text{ m/s}^2$$

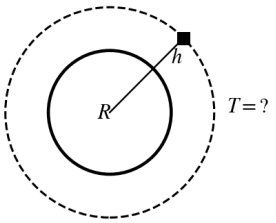
The person's mass cancels ($mg = GMm/r^2$); it's extra information.

Why the other choices are wrong:

- b) 12.6 m/s²: No standard mistake gives this value; it is a spacing choice.
- c) 17.6 m/s²: No standard mistake gives this value; it is a spacing choice.
- d) 24.6 m/s²: No standard mistake gives this value; it is a spacing choice.
- e) 34.4 m/s²: No standard mistake gives this value; it is a spacing choice.

10. Satellite period — e) 101 min

Source: Ch 13 notes p. 2/10 (satellite orbiting a planet, $T^2 = 4\pi^2 r^3/GM$, $r = R + h$; $h = 400$ km $\to$ 92.4 min).
Previous-exam match: 2015 Exam 3 #10 (altitude 10^8 m: 3.99 days $\to$ d) 4 days), new numbers
Notecard: Orbit
What it tests: $T^2 = 4\pi^2 r^3/GM$ with $r = R + h$.



Step 1: gravity supplies the centripetal force (notes p. 2/10):

$$\frac{GMm}{r^2} = \frac{mv^2}{r}, \qquad v = \frac{2\pi r}{T} \;\Rightarrow\; T^2 = \frac{4\pi^2 r^3}{GM}$$

The satellite's mass m cancels.

Step 2: orbit radius from Earth's center:

$$r = \underbrace{6.37 \times 10^6}_{R} + \underbrace{8 \times 10^5}_{h} = 7.170 \times 10^6 \text{ m}$$

Step 3: plug in:

$$T = \sqrt{\frac{4\pi^2 (7.170 \times 10^6)^3}{\underbrace{(6.67 \times 10^{-11})(5.98 \times 10^{24})}_{GM_{earth}}}} = 6{,}040 \text{ s}$$

Step 4: convert: 6,040 s = 101 min.

Notes example: $h = 400$ km gives $T = 5546$ s = 92.4 min (low Earth orbit).

Why the other choices are wrong:

- a) 3.75 min: uses h instead of $r = R + h$.
- b) 16.0 min: drops the $4\pi^2$ (forgets the 2π in $v = 2\pi r/T$).

- c) 50.3 min: uses π instead of 2π.
- d) 84.3 min: uses R_{earth} (an orbit skimming the surface).

11. Height reached from the launch speed — a) 493 km

Source: Ch 13 notes p. 7/10 and p. 8/10 (rocket launch $v_i = 1000$ m/s, $h = \frac{2GMR}{2GM - v_i^2 R} - R$).
Previous-exam match: none (lecture-notes example)
Notecard: Rocket
What it tests: energy conservation with $U = -GMm/r$, solved for h.

Step 1: conservation of energy (notes p. 7/10), launch at $r = R$ with speed v_i, top at r with speed 0:

$$-\frac{GMm}{R} + \frac{1}{2}mv_i^2 = -\frac{GMm}{r} \;\Rightarrow\; r = \frac{1}{\frac{1}{R} - \frac{v_i^2}{2GM}} = \frac{2GMR}{2GM - v_i^2 R}$$

Step 2: plug in ($2GM = 2(6.67 \times 10^{-11})(5.98 \times 10^{24}) = 7.977 \times 10^{14}$):

$$r = \frac{\underbrace{(7.977 \times 10^{14})}_{2GM}\underbrace{(6.37 \times 10^{6})}_{R}}{7.977 \times 10^{14} - \underbrace{(3{,}000)^2}_{v_i^2}(6.37 \times 10^{6})} = 6.8632 \times 10^{6} \text{ m}$$

Step 3: height above the surface:

$$h = r - R = 6.8632 \times 10^{6} - 6.37 \times 10^{6} = 4.93 \times 10^{5} \text{ m} = 493 \text{ km}$$

Keep at least 4 significant figures until the subtraction: r and R are close, so early rounding changes the answer. The notes' $v_i = 1000$ m/s example shows $h = 50.73$ km; carried through without rounding, it is 51.4 km.

Why the other choices are wrong:

- b) 918 km: uses v^2/g.
- c) 1,070 km: forgets the $\frac{1}{2}$ in $\frac{1}{2}mv^2$.
- d) 6,860 km: gives r, the distance from Earth's center; forgets to subtract R.
- e) 9,600 km: No standard mistake gives this value; it is a spacing choice.

12. Launch speed to reach a height — b) 5.47 km/s

Source: Ch 13 notes p. 7/10 ($U = -GMm/r$, energy conservation) and p. 8/10 ("given a target altitude $h = 1000$ km, what v to launch?" 4119 m/s).
Previous-exam match: 2015 Exam 3 #9 (1000 km), new numbers
Notecard: Rocket
What it tests: energy conservation with $U = -GMm/r$, solved for v.

Step 1: conservation of energy (notes p. 7–8/10). At launch $r = R$, speed v; at the top $r = R + h$, speed 0. Gravitational PE is $U = -GMm/r$:

$$-\frac{GMm}{R} + \frac{1}{2}mv^2 = -\frac{GMm}{r} \;\Rightarrow\; v = \sqrt{2GM\Big(\frac{1}{R} - \frac{1}{r}\Big)}$$

Step 2: top of the flight: $r = R + h = 6.37 \times 10^6 + 2 \times 10^6 = 8.370 \times 10^6$ m.

Step 3: plug in:

$$v = \sqrt{2\,\underbrace{(6.67 \times 10^{-11})(5.98 \times 10^{24})}_{GM_{earth}}\Big(\underbrace{\frac{1}{6.37 \times 10^6}}_{1/R} - \underbrace{\frac{1}{8.370 \times 10^6}}_{1/r}\Big)} = 5{,}470 \text{ m/s} = 5.47 \text{ km/s}$$

Notes example: $h = 1000$ km needs $v = 4119$ m/s.

Why the other choices are wrong:

- a) 3.87 km/s: forgets the 2 from $\frac{1}{2}mv^2$.
- c) 6.26 km/s: uses mgh (constant g), which is wrong this far up.
- d) 11.2 km/s: is the escape speed.
- e) 15.7 km/s: No standard mistake gives this value; it is a spacing choice.

13. Escape speed — c) 5.01 km/s

Source: Ch 13 notes p. 9/10 ($v_{escape} = \sqrt{2GM/R} = 11.2$ km/s for Earth).
Previous-exam match: none (lecture-notes example)
Notecard: Escape
What it tests: $v_{esc} = \sqrt{2GM/R}$.

Step 1: energy conservation (notes p. 9/10). To escape, the object just reaches $r = \infty$ with $v = 0$, where $U = 0$ and $K = 0$:

$$-\frac{GMm}{R} + \frac{mv^2}{2} = 0 + 0 \;\Rightarrow\; v_{esc} = \sqrt{\frac{2GM}{R}}$$

Step 2: plug in:

$$v_{esc} = \sqrt{\frac{2\,\underbrace{(6.67\times10^{-11})}_{G}\,\underbrace{(6.4\times10^{23})}_{M}}{\underbrace{3.4\times10^{6}}_{R}}} = 5{,}011 \text{ m/s} = 5.01 \text{ km/s}$$

For Earth the notes get 11,181 m/s, about 11.2 km/s.

Why the other choices are wrong:

- a) 2.53 km/s: No standard mistake gives this value; it is a spacing choice.
- b) 3.54 km/s: forgets the 2 (that is the orbital speed at the surface).
- d) 10.0 km/s: doubles the answer.
- e) 14.0 km/s: No standard mistake gives this value; it is a spacing choice.

14. Equal-pull point — d) 89.9%

Source: 2015 Exam 3 #13, solved with $F = GMm/r^2$ from Ch 13 notes p. 1/10.
Previous-exam match: 2015 Exam 3 #13 (moon 7×10^{22} kg, 400,000 km: 90.2% → b) 90%), new numbers
Notecard: Equal pull
What it tests: setting two inverse-square pulls equal: $x/d = k/(1+k)$, $k = \sqrt{M_E/M_b}$.

Step 1: set the two pulls equal with Newton's law (Ch 13 notes p. 1/10). Let x be the distance from Earth's center, so the distance to the moon is $d - x$. The test mass m cancels:

$$\frac{GM_E m}{x^2} = \frac{GM_b m}{(d-x)^2} \;\Rightarrow\; \frac{x}{d-x} = \sqrt{\frac{M_E}{M_b}}$$

Step 2: the square root:

$$k = \sqrt{\frac{\overbrace{5.98 \times 10^{24}}^{M_E}}{\underbrace{7.475 \times 10^{22}}_{M_b}}} = 8.944$$

Step 3: solve for the fraction: $x = k(d - x)$, so

$$\frac{x}{d} = \frac{k}{1+k} = \frac{8.944}{9.944} = 0.8994 = 89.9\%$$

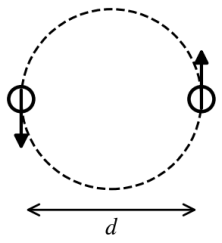
The distance doesn't matter for the percentage; only the mass ratio does.

Why the other choices are wrong:

- a) 10.1%: gives the distance from the moon's center.
- b) 50.0%: assumes the halfway point.
- c) 81.2%: uses a cube root.
- e) 98.8%: uses the mass ratio without the square root.

15. Binary stars, period — c) 87.2 hours

Source: Ch 13 notes p. 4/10 (two equal-mass stars, $T^2 = 16\pi^2 r^3/GM$; "separation btwn the masses is $2r$ (not r)").
Previous-exam match: 2015 Exam 3 #11 (10^{31} kg each, $d = 10^8$ m: 2.9 min → a) 3 minutes), new numbers
Notecard: Binary stars
What it tests: $T^2 = 16\pi^2 r^3/GM$ with r = half the separation.



Step 1: the force uses the separation, the circle uses r (notes p. 4/10). Each star moves on a circle of radius r about the center; the distance between the stars is $2r$:

$$\frac{GM^2}{(2r)^2} = \frac{Mv^2}{r}, \qquad v = \frac{2\pi r}{T} \;\Rightarrow\; T^2 = \frac{16\pi^2 r^3}{GM}$$

Step 2: the orbit radius: $r = d/2 = 5 \times 10^9$ m.

Step 3: plug in:

$$T = \sqrt{\frac{16\pi^2(5\times 10^9)^3}{\underbrace{(6.67\times 10^{-11})}_{G}\underbrace{(3\times 10^{30})}_{M}}} = 3.141 \times 10^5 \text{ s} = 87.2 \text{ hours}$$

Notes example: $M = 2 \times 10^{30}$ kg, $r = 6 \times 10^{10}$ m gives 185 days.

Why the other choices are wrong:

- a) 43.6 hours: uses the single-planet $4\pi^2$ formula (forgets that the separation is $2r$ in the force).
- b) 61.7 hours: uses $2M$ for the mass.
- d) 123 hours: uses the single-planet formula with $r = d$.
- e) 247 hours: uses the separation d as the orbit radius (the notes warn: separation is $2r$, not r).